SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 6 – Short Answer (120 Questions)

Course Code:	ITA0332	**Course Title:**	Mobile computing
CO Assessed:	CO2 – Manipulation (BL4)	**Assessment:**	Assessment Tool 6 – Short Answer
Weightage:	30 % of CIA	**Total Marks:**	20 (Answer ANY TEN)
CO3 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	CH.V.Ramayogi	**Reg. No.:**	192424319
Section / Batch:	SLOT-A/B.E-AIDS 2024-28	**Date:**	4/11/2026

Code of Conduct

I CH.V.Ramayogi/192424319(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Ch.V.Ramayogi

Instructions

- This test contains 120 questions (ANSWER ANY TEN), Total: 20 Marks.
- No calculators or electronic devices permitted.
- Duration: 30 minutes.

ASSESSMENT TOOL 6 – SHORT ANSWERS

Question 1

1. Hybrid scenario monitoring scenario where protocol stack suffers from interference issues, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 2

2. High-speed deployment analysis scenario where protocol stack shows high latency variation, examine system behavior in detail. Propose enhancements with justification.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Since the deployment is high-speed, validate the changes while users move between cells.

Question 3

3. Hybrid environment evaluation scenario where radio interface shows unstable throughput, examine system behavior in detail. Design an optimized communication approach.

Answer: The radio interface carries control and user data between a mobile device and base station. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 4

4. Hybrid scenario assessment scenario where 3G shows unstable throughput, examine system behavior in detail. Identify key factors and propose redesign.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 5

5. Secure framework migration scenario where 4G suffers from interference issues, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Use carrier aggregation, load balancing and small cells where demand is high. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 6

6. Distributed platform monitoring scenario where spread spectrum experiences frequent handoff failures, examine system behavior in detail. Investigate causes and model improvements.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. In a distributed platform, check both radio performance and the links connecting network sites.

Question 7

7. Rural deployment evaluation scenario where 2G demonstrates poor QoS, examine system behavior in detail. Identify key factors and propose redesign.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 8

8. Wide-area system analysis scenario where radio interface shows high latency variation, examine system behavior in detail. Design an optimized communication approach.

Answer: The radio interface carries control and user data between a mobile device and base station. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For a wide-area system, verify performance across distant cells and transport links.

Question 9

9. Wireless platform evaluation scenario where 3G shows high latency variation, examine system behavior in detail. Develop a robust optimization technique.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 10

10. Next-gen system control scenario where multiple access has energy inefficiency, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Energy inefficiency increases battery drain or network power use through

unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a next-generation network, ensure the change supports growing traffic and future services.

Question 11

11. Legacy architecture migration scenario where HSCSD suffers from interference issues, examine system behavior in detail. Design an optimized communication approach.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 12

12. Next-gen network analysis scenario where 5G reveals packet loss anomalies, examine system behavior in detail. Develop a robust optimization technique.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. For a next-generation network, ensure the change supports growing traffic and future services.

Question 13

13. Secure network analysis scenario where multiple access encounters scalability challenges, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 14

14. Secure architecture evaluation scenario where 2G suffers from interference issues, examine system behavior in detail. Propose enhancements with justification.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 15

15. Mobile infrastructure design scenario where 5G has energy inefficiency, examine system behavior in detail. Propose enhancements with justification.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. In a mobile infrastructure, test both stationary and moving devices.

Question 16

16. Distributed deployment design scenario where 3G experiences frequent handoff failures, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. In a distributed platform, check both radio performance and the links connecting network sites.

Question 17

17. Next-gen setup assessment scenario where 4G encounters scalability challenges, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Use carrier aggregation, load balancing and small cells where demand is high. For a next-generation network, ensure the change supports growing traffic and future services.

Question 18

18. Legacy network migration scenario where GPRS demonstrates poor QoS, examine system behavior in detail. Develop a robust optimization technique.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 19

19. Hybrid setup implementation scenario where radio interface encounters scalability challenges, examine system behavior in detail. Propose enhancements with justification.

Answer: The radio interface carries control and user data between a mobile device and base station. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 20

20. Massive deployment integration scenario where spread spectrum exposes security weaknesses, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. Test the proposed change under normal and peak traffic before deployment.

Question 21

21. Mobile system integration scenario where 1G shows unstable throughput, examine system behavior in detail. Propose enhancements with justification.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Test the proposed change under normal and peak traffic before deployment.

Question 22

22. Rural network design scenario where 3G experiences frequent handoff failures, examine system behavior in detail. Identify key factors and propose redesign.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 23

23. Next-gen deployment control scenario where multiple access faces unexpected congestion, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a next-generation network, ensure the change supports growing traffic and future services.

Question 24

24. Urban system assessment scenario where HSCSD exposes security weaknesses, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. Test the proposed change under normal and peak traffic before deployment.

Question 25

25. Low-latency architecture assessment scenario where GPRS demonstrates poor QoS, examine system behavior in detail. Identify key factors and propose redesign.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Test the proposed change under normal and peak traffic before deployment.

Question 26

26. Intelligent network evaluation scenario where multiple access demonstrates poor QoS, examine system behavior in detail. Identify key factors and propose redesign.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. Test the proposed change under normal and peak traffic before deployment.

Question 27

27. Adaptive environment assessment scenario where radio interface experiences frequent handoff failures, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: The radio interface carries control and user data between a mobile device and base station. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Test the proposed change under normal and peak traffic before deployment.

Question 28

28. Wireless framework implementation scenario where GPRS has energy inefficiency, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 29

29. Massive infrastructure analysis scenario where protocol stack exposes security weaknesses, examine system behavior in detail. Propose enhancements with justification.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 30

30. Dense setup migration scenario where protocol stack reveals packet loss anomalies, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 31

31. Distributed setup monitoring scenario where 4G suffers from interference issues, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Use carrier aggregation, load balancing and small cells where demand is high. In a distributed platform, check both radio performance and the links connecting network sites.

Question 32

32. Legacy architecture design scenario where 1G reveals packet loss anomalies, examine system behavior in detail. Propose enhancements with justification.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 33

33. Rural setup control scenario where protocol stack reveals packet loss anomalies, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 34

34. Legacy setup migration scenario where multiple access suffers from interference issues, examine system behavior in detail. Propose enhancements with justification.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 35

35. Massive setup optimization scenario where spread spectrum exposes security weaknesses, examine system behavior in detail. Develop a robust optimization technique.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. Test the proposed change under normal and peak traffic before deployment.

Question 36

36. Mobile setup assessment scenario where 3G shows unstable throughput, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. Test the proposed change under normal and peak traffic before deployment.

Question 37

37. Secure architecture migration scenario where GPRS has energy inefficiency, examine system behavior in detail. Develop a robust optimization technique.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 38

38. Next-gen platform optimization scenario where radio interface experiences frequent handoff failures, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: The radio interface carries control and user data between a mobile device and base station. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target

coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For a next-generation network, ensure the change supports growing traffic and future services.

Question 39

39. Wide-area setup monitoring scenario where GPRS shows unstable throughput, examine system behavior in detail. Propose enhancements with justification.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. For a wide-area system, verify performance across distant cells and transport links.

Question 40

40. Adaptive deployment integration scenario where 2G faces unexpected congestion, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. Test the proposed change under normal and peak traffic before deployment.

Question 41

41. Urban framework optimization scenario where protocol stack shows high latency variation, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 42

42. High-speed deployment evaluation scenario where 3G exposes security weaknesses, examine system behavior in detail. Investigate causes and model improvements.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. Since the deployment is high-speed, validate the changes while users move between cells.

Question 43

43. Intelligent framework optimization scenario where 1G exposes security weaknesses, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Test the proposed change under normal and peak traffic before deployment.

Question 44

44. Adaptive platform integration scenario where 3G experiences frequent handoff failures, examine system behavior in detail. Design an optimized communication approach.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. Test the proposed change under normal and peak traffic before deployment.

Question 45

45. Low-latency system integration scenario where 2G suffers from interference issues, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. Test the proposed change under normal and peak traffic before deployment.

Question 46

46. Secure setup integration scenario where 3G suffers from interference issues, examine system behavior in detail. Design an optimized communication approach.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 47

47. Distributed scenario optimization scenario where protocol stack demonstrates poor QoS, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. In a distributed platform, check both radio performance and the links connecting network sites.

Question 48

48. Distributed scenario control scenario where 4G suffers from interference issues, examine system behavior in detail. Investigate causes and model improvements.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Use carrier aggregation, load balancing and small cells where demand is high. In a distributed platform, check both radio performance and the links connecting network sites.

Question 49

49. Dense architecture evaluation scenario where multiple access suffers from interference issues, examine system behavior in detail. Identify key factors and propose redesign.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. Test the proposed change under normal and peak traffic before deployment.

Question 50

50. Wide-area network design scenario where HSCSD faces unexpected congestion, examine system behavior in detail. Develop a robust optimization technique.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. For a wide-area system, verify performance across distant cells and transport links.

Question 51

51. Wireless infrastructure optimization scenario where 1G faces unexpected congestion, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 52

52. Legacy environment design scenario where 2G demonstrates poor QoS, examine system behavior in detail. Develop a robust optimization technique.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 53

53. Advanced system evaluation scenario where spread spectrum encounters scalability challenges, examine system behavior in detail. Develop a robust optimization technique.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. Test the proposed change under normal and peak traffic before deployment.

Question 54

54. Secure system evaluation scenario where 1G exposes security weaknesses, examine system behavior in detail. Investigate causes and model improvements.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 55

55. Autonomous framework optimization scenario where 5G demonstrates poor QoS, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 56

56. Dynamic setup evaluation scenario where GPRS encounters scalability challenges, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. In a dynamic platform, review performance during peak and off-peak traffic.

Question 57

57. Wide-area scenario design scenario where 4G has energy inefficiency, examine system behavior in detail. Identify key factors and propose redesign.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Use carrier aggregation, load balancing and small cells where demand is high. For a wide-area system, verify performance across distant cells and transport links.

Question 58

58. Rural network optimization scenario where 5G experiences frequent handoff failures, examine system behavior in detail. Identify key factors and propose redesign.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 59

59. Secure system design scenario where GPRS encounters scalability challenges, examine system behavior in detail. Investigate causes and model improvements.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 60

60. Urban infrastructure design scenario where 3G exposes security weaknesses, examine system behavior in detail. Develop a robust optimization technique.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. Test the proposed change under normal and peak traffic before deployment.

Question 61

61. Next-gen infrastructure analysis scenario where 4G experiences frequent handoff failures, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Use carrier aggregation, load balancing and small cells where demand is high. For a next-generation network, ensure the change supports growing traffic and future services.

Question 62

62. Autonomous system analysis scenario where HSCSD has energy inefficiency, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 63

63. Dense system monitoring scenario where 5G faces unexpected congestion, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. Test the proposed change under normal and peak traffic before deployment.

Question 64

64. Autonomous network design scenario where 2G experiences frequent handoff failures, examine system behavior in detail. Propose enhancements with justification.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 65

65. Hybrid framework migration scenario where protocol stack demonstrates poor QoS, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 66

66. Mobile system analysis scenario where multiple access shows high latency variation, examine system behavior in detail. Propose enhancements with justification.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. Test the proposed change under normal and peak traffic before deployment.

Question 67

67. Distributed framework migration scenario where 5G exposes security weaknesses, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. In a distributed platform, check both radio performance and the links connecting network sites.

Question 68

68. Urban network control scenario where protocol stack has energy inefficiency, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 69

69. Wireless scenario monitoring scenario where 2G encounters scalability challenges, examine system behavior in detail. Identify key factors and propose redesign.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 70

70. Advanced environment evaluation scenario where 4G shows unstable throughput, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Use carrier aggregation, load balancing and small cells where demand is high. Test the proposed change under normal and peak traffic before deployment.

Question 71

71. Low-latency infrastructure control scenario where 1G suffers from interference issues, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Test the proposed change under normal and peak traffic before deployment.

Question 72

72. Legacy deployment assessment scenario where 3G exposes security weaknesses, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 73

73. Secure deployment optimization scenario where 1G demonstrates poor QoS, examine system behavior in detail. Investigate causes and model improvements.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 74

74. Massive environment optimization scenario where radio interface suffers from interference issues, examine system behavior in detail. Identify key factors and propose redesign.

Answer: The radio interface carries control and user data between a mobile device and base station. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Test the proposed change under normal and peak traffic before deployment.

Question 75

75. Adaptive infrastructure control scenario where multiple access demonstrates poor QoS, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. Test the proposed change under normal and peak traffic before deployment.

Question 76

76. High-speed scenario optimization scenario where radio interface demonstrates poor QoS, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: The radio interface carries control and user data between a mobile device and base station. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Since the deployment is high-speed, validate the changes while users move between cells.

Question 77

77. Next-gen scenario migration scenario where 4G encounters scalability challenges, examine system behavior in detail. Design an optimized communication approach.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Use carrier aggregation, load balancing and small cells where demand is high. For a next-generation network, ensure the change supports growing traffic and future services.

Question 78

78. Next-gen deployment analysis scenario where multiple access shows high latency variation, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a next-generation network, ensure the change supports growing traffic and future services.

Question 79

79. Next-gen setup design scenario where 1G shows high latency variation, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. For a next-generation network, ensure the change supports growing traffic and future services.

Question 80

80. Next-gen scenario optimization scenario where 4G encounters scalability challenges, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Use carrier aggregation, load balancing and small cells where demand is high. For a next-generation network, ensure the change supports growing traffic and future services.

Question 81

81. Autonomous environment control scenario where spread spectrum suffers from interference issues, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 82

82. Massive system evaluation scenario where GPRS exposes security weaknesses, examine system behavior in detail. Propose enhancements with justification.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Test the proposed change under normal and peak traffic before deployment.

Question 83

83. Wireless deployment evaluation scenario where multiple access experiences frequent handoff failures, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 84

84. Secure environment monitoring scenario where spread spectrum shows high latency variation, examine system behavior in detail. Develop a robust optimization technique.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 85

85. Hybrid system monitoring scenario where GPRS shows unstable throughput, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 86

86. Dense environment optimization scenario where 2G shows high latency variation, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. Test the proposed change under normal and peak traffic before deployment.

Question 87

87. Rural infrastructure integration scenario where spread spectrum demonstrates poor QoS, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Poor QoS appears as low speed, high delay, packet loss or call

drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 88

88. Critical network implementation scenario where radio interface has energy inefficiency, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: The radio interface carries control and user data between a mobile device and base station. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For a critical environment, give priority to reliable service and secure communication.

Question 89

89. Mobile network evaluation scenario where HSCSD faces unexpected congestion, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. Test the proposed change under normal and peak traffic before deployment.

Question 90

90. Low-latency network analysis scenario where GPRS exposes security weaknesses, examine system behavior in detail. Develop a robust optimization technique.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Test the proposed change under normal and peak traffic before deployment.

Question 91

91. Secure system assessment scenario where 2G faces unexpected congestion, examine system behavior in detail. Investigate causes and model improvements.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 92

92. Adaptive infrastructure migration scenario where HSCSD shows unstable throughput, examine system behavior in detail. Develop a robust optimization technique.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. Test the proposed change under normal and peak traffic before deployment.

Question 93

93. Secure system monitoring scenario where 5G reveals packet loss anomalies, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 94

94. Wireless architecture control scenario where radio interface has energy inefficiency, examine system behavior in detail. Investigate causes and model improvements.

Answer: The radio interface carries control and user data between a mobile device and base station. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 95

95. Advanced architecture migration scenario where 1G has energy inefficiency, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Test the proposed change under normal and peak traffic before deployment.

Question 96

96. Hybrid network assessment scenario where 1G has energy inefficiency, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 97

97. Massive framework monitoring scenario where GPRS has energy inefficiency, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Test the proposed change under normal and peak traffic before deployment.

Question 98

98. Adaptive environment analysis scenario where 4G faces unexpected congestion, examine system behavior in detail. Investigate causes and model improvements.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Use carrier aggregation, load balancing and small cells where demand is high. Test the proposed change under normal and peak traffic before deployment.

Question 99

99. Distributed environment analysis scenario where 4G shows unstable throughput, examine system behavior in detail. Design an optimized communication approach.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Use carrier aggregation, load balancing and small cells where demand is high. In a distributed platform, check both radio performance and the links connecting network sites.

Question 100

100. Distributed network migration scenario where 2G shows high latency variation, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. In a distributed platform, check both radio performance and the links connecting network sites.

Question 101

101. High-speed network assessment scenario where GPRS reveals packet loss anomalies, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Since the deployment is high-speed, validate the changes while users move between cells.

Question 102

102. Advanced infrastructure design scenario where protocol stack suffers from interference issues, examine system behavior in detail. Investigate causes and model improvements.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 103

103. Wireless environment design scenario where HSCSD demonstrates poor QoS, examine system behavior in detail. Propose enhancements with justification.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or

weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 104

104. Mobile environment evaluation scenario where radio interface has energy inefficiency, examine system behavior in detail. Propose enhancements with justification.

Answer: The radio interface carries control and user data between a mobile device and base station. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Test the proposed change under normal and peak traffic before deployment.

Question 105

105. Critical environment assessment scenario where radio interface suffers from interference issues, examine system behavior in detail. Propose enhancements with justification.

Answer: The radio interface carries control and user data between a mobile device and base station. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For a critical environment, give priority to reliable service and secure communication.

Question 106

106. Massive framework integration scenario where 5G experiences frequent handoff failures, examine system behavior in detail. Propose enhancements with justification.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. Test the proposed change under normal and peak traffic before deployment.

Question 107

107. Low-latency system implementation scenario where 5G reveals packet loss anomalies, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. Test the proposed change under normal and peak traffic before deployment.

Question 108

108. Autonomous system assessment scenario where radio interface shows unstable throughput, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: The radio interface carries control and user data between a mobile device and base station. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 109

109. High-speed infrastructure integration scenario where 4G has energy inefficiency, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Use carrier aggregation, load balancing and small cells where demand is high. Since the deployment is high-speed, validate the changes while users move between cells.

Question 110

110. Advanced setup assessment scenario where spread spectrum has energy inefficiency, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. Test the proposed change under normal and peak traffic before deployment.

Question 111

111. Wireless architecture implementation scenario where spread spectrum has energy inefficiency, examine system behavior in detail. Investigate causes and model improvements.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 112

112. Wide-area framework evaluation scenario where 2G suffers from interference issues, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For a wide-area system, verify performance across distant cells and transport links.

Question 113

113. Wide-area system assessment scenario where 3G shows high latency variation, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. For a wide-area system, verify performance across distant cells and transport links.

Question 114

114. Dense deployment migration scenario where radio interface shows unstable throughput, examine system behavior in detail. Investigate causes and model improvements.

Answer: The radio interface carries control and user data between a mobile device and base station. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Test the proposed change under normal and peak traffic before deployment.

Question 115

115. Urban network analysis scenario where HSCSD reveals packet loss anomalies, examine system behavior in detail. Investigate causes and model improvements.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. Test the proposed change under normal and peak traffic before deployment.

Question 116

116. Rural environment migration scenario where protocol stack has energy inefficiency, examine system behavior in detail. Investigate causes and model improvements.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Energy inefficiency increases battery drain or network power use through

unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 117

117. Next-gen infrastructure evaluation scenario where multiple access shows high latency variation, examine system behavior in detail. Propose enhancements with justification.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a next-generation network, ensure the change supports growing traffic and future services.

Question 118

118. High-speed infrastructure evaluation scenario where 2G encounters scalability challenges, examine system behavior in detail. Develop a robust optimization technique.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. Since the deployment is high-speed, validate the changes while users move between cells.

Question 119

119. Dynamic environment monitoring scenario where HSCSD exposes security weaknesses, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. In a dynamic platform, review performance during peak and off-peak traffic.

Question 120

120. Intelligent scenario optimization scenario where protocol stack reveals packet loss anomalies, examine system behavior in detail. Investigate causes and model improvements.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 121

121. Distributed infrastructure migration scenario where GPRS suffers from interference issues, examine system behavior in detail. Investigate causes and model improvements.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. In a distributed platform, check both radio performance and the links connecting network sites.

Question 122

122. Urban infrastructure assessment scenario where 5G shows unstable throughput, examine system behavior in detail. Investigate causes and model improvements.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. Test the proposed change under normal and peak traffic before deployment.

Question 123

123. Critical network integration scenario where HSCSD shows high latency variation, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. For a critical environment, give priority to reliable service and secure communication.

Question 124

124. Critical framework control scenario where 2G has energy inefficiency, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For a critical environment, give priority to reliable service and secure communication.

Question 125

125. Dense architecture analysis scenario where protocol stack demonstrates poor QoS, examine system behavior in detail. Propose enhancements with justification.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 126

126. Low-latency framework control scenario where multiple access experiences frequent handoff failures, examine system behavior in detail. Design an optimized communication approach.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. Test the proposed change under normal and peak traffic before deployment.

Question 127

127. Next-gen environment control scenario where multiple access shows high latency variation, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a next-generation network, ensure the change supports growing traffic and future services.

Question 128

128. Intelligent setup optimization scenario where radio interface has energy inefficiency, examine system behavior in detail. Propose enhancements with justification.

Answer: The radio interface carries control and user data between a mobile device and base station. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Test the proposed change under normal and peak traffic before deployment.

Question 129

129. Rural setup monitoring scenario where radio interface exposes security weaknesses, examine system behavior in detail. Design an optimized communication approach.

Answer: The radio interface carries control and user data between a mobile device and base station. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 130

130. Hybrid architecture monitoring scenario where protocol stack has energy inefficiency, examine system behavior in detail. Investigate causes and model improvements.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 131

131. Mobile architecture optimization scenario where HSCSD demonstrates poor QoS, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. Test the proposed change under normal and peak traffic before deployment.

Question 132

132. Dense framework optimization scenario where protocol stack suffers from interference issues, examine system behavior in detail. Develop a robust optimization technique.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 133

133. Low-latency system monitoring scenario where 1G experiences frequent handoff failures, examine system behavior in detail. Design an optimized communication approach.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Test the proposed change under normal and peak traffic before deployment.

Question 134

134. Mobile scenario design scenario where HSCSD demonstrates poor QoS, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. Test the proposed change under normal and peak traffic before deployment.

Question 135

135. Low-latency system optimization scenario where 5G faces unexpected congestion, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. Test the proposed change under normal and peak traffic before deployment.

Question 136

136. Low-latency environment control scenario where 4G encounters scalability challenges, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Use carrier aggregation, load balancing and small cells where demand is high. Test the proposed change under normal and peak traffic before deployment.

Question 137

137. Next-gen architecture optimization scenario where radio interface exposes security weaknesses, examine system behavior in detail. Develop a robust optimization technique.

Answer: The radio interface carries control and user data between a mobile device and base station. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For a next-generation network, ensure the change supports growing traffic and future services.

Question 138

138. Low-latency environment assessment scenario where GPRS experiences frequent handoff failures, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Test the proposed change under normal and peak traffic before deployment.

Question 139

139. Secure setup analysis scenario where radio interface has energy inefficiency, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: The radio interface carries control and user data between a mobile device and base station. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 140

140. Wide-area network migration scenario where radio interface encounters scalability challenges, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: The radio interface carries control and user data between a mobile device and base station. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For a wide-area system, verify performance across distant cells and transport links.

Question 141

141. Distributed framework design scenario where protocol stack demonstrates poor QoS, examine system behavior in detail. Design an optimized communication approach.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. In a distributed platform, check both radio performance and the links connecting network sites.

Question 142

142. High-speed scenario design scenario where spread spectrum faces unexpected congestion, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. Since the deployment is high-speed, validate the changes while users move between cells.

Question 143

143. High-speed infrastructure analysis scenario where 4G experiences frequent handoff failures, examine system behavior in detail. Investigate causes and model improvements.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Use carrier aggregation, load balancing and small cells where demand is high. Since the deployment is high-speed, validate the changes while users move between cells.

Question 144

144. Legacy network control scenario where GPRS reveals packet loss anomalies, examine system behavior in detail. Design an optimized communication approach.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 145

145. High-speed platform optimization scenario where 2G has energy inefficiency, examine system behavior in detail. Propose enhancements with justification.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. Since the deployment is high-speed, validate the changes while users move between cells.

Question 146

146. Autonomous platform implementation scenario where 2G experiences frequent handoff failures, examine system behavior in detail. Investigate causes and model improvements.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 147

147. Wide-area scenario assessment scenario where multiple access exposes security weaknesses, examine system behavior in detail. Develop a robust optimization technique.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a wide-area system, verify performance across distant cells and transport links.

Question 148

148. Massive environment assessment scenario where 3G has energy inefficiency, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. Test the proposed change under normal and peak traffic before deployment.

Question 149

149. Adaptive architecture migration scenario where protocol stack has energy inefficiency, examine system behavior in detail. Design an optimized communication approach.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 150

150. Legacy deployment implementation scenario where HSCSD demonstrates poor QoS, examine system behavior in detail. Investigate causes and model improvements.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 151

151. Mobile deployment design scenario where 3G exposes security weaknesses, examine system behavior in detail. Design an optimized communication approach.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. Test the proposed change under normal and peak traffic before deployment.

Question 152

152. Rural infrastructure implementation scenario where 2G shows unstable throughput, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Unstable throughput can result from weak signal, fading, interference, cell congestion or backhaul limits. Measure SINR and cell load, improve coverage, balance traffic and add carrier or backhaul capacity where required. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 153

153. Mobile platform integration scenario where 2G suffers from interference issues, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections.

Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. Test the proposed change under normal and peak traffic before deployment.

Question 154

154. Next-gen framework integration scenario where multiple access reveals packet loss anomalies, examine system behavior in detail. Identify key factors and propose redesign.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a next-generation network, ensure the change supports growing traffic and future services.

Question 155

155. Wireless framework migration scenario where GPRS faces unexpected congestion, examine system behavior in detail. Investigate causes and model improvements.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 156

156. Wide-area system control scenario where 2G suffers from interference issues, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For a wide-area system, verify performance across distant cells and transport links.

Question 157

157. Dense platform analysis scenario where 1G demonstrates poor QoS, examine system behavior in detail. Investigate causes and model improvements.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Test the proposed change under normal and peak traffic before deployment.

Question 158

158. Autonomous system integration scenario where radio interface demonstrates poor QoS, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: The radio interface carries control and user data between a mobile device and base station. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 159

159. Next-gen network control scenario where multiple access suffers from interference issues, examine system behavior in detail. Develop a robust optimization technique.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a next-generation network, ensure the change supports growing traffic and future services.

Question 160

160. Distributed deployment migration scenario where GPRS faces unexpected congestion, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. In a distributed platform, check both radio performance and the links connecting network sites.

Question 161

161. Advanced network migration scenario where GPRS encounters scalability challenges, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Test the proposed change under normal and peak traffic before deployment.

Question 162

162. Rural scenario design scenario where 2G shows high latency variation, examine system behavior in detail. Design an optimized communication approach.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 163

163. Secure framework evaluation scenario where multiple access demonstrates poor QoS, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to

congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. For a secure framework, retain strong authentication and protect signalling during optimization.

Question 164

164. Adaptive network control scenario where 4G shows high latency variation, examine system behavior in detail. Investigate causes and model improvements.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Use carrier aggregation, load balancing and small cells where demand is high. Test the proposed change under normal and peak traffic before deployment.

Question 165

165. High-speed platform design scenario where 4G exposes security weaknesses, examine system behavior in detail. Investigate causes and model improvements.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Use carrier aggregation, load balancing and small cells where demand is high. Since the deployment is high-speed, validate the changes while users move between cells.

Question 166

166. Mobile setup implementation scenario where spread spectrum faces unexpected congestion, examine system behavior in detail. Propose enhancements with justification.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. Test the proposed change under normal and peak traffic before deployment.

Question 167

167. Dense scenario integration scenario where 4G suffers from interference issues, examine system behavior in detail. Design an optimized communication approach.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Use carrier aggregation, load balancing and small cells where demand is high. Test the proposed change under normal and peak traffic before deployment.

Question 168

168. Hybrid network optimization scenario where 2G demonstrates poor QoS, examine system behavior in detail. Identify key factors and propose redesign.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 169

169. Urban deployment analysis scenario where protocol stack reveals packet loss anomalies, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. Test the proposed change under normal and peak traffic before deployment.

Question 170

170. Massive system design scenario where spread spectrum reveals packet loss anomalies, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. Test the proposed change under normal and peak traffic before deployment.

Question 171

171. Legacy platform assessment scenario where spread spectrum shows high latency variation, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. In a legacy architecture, use compatible improvements while planning migration from obsolete technology.

Question 172

172. Adaptive deployment assessment scenario where 4G suffers from interference issues, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Use carrier aggregation, load balancing and small cells where demand is high. Test the proposed change under normal and peak traffic before deployment.

Question 173

173. Wireless architecture migration scenario where protocol stack exposes security weaknesses, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. In a wireless platform, test the solution under realistic radio conditions and user load.

Question 174

174. Massive network analysis scenario where 3G exposes security weaknesses, examine system behavior in detail. Investigate causes and model improvements.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. Test the proposed change under normal and peak traffic before deployment.

Question 175

175. Wide-area infrastructure design scenario where 5G suffers from interference issues, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. For a wide-area system, verify performance across distant cells and transport links.

Question 176

176. Autonomous network optimization scenario where spread spectrum experiences frequent handoff failures, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 177

177. Adaptive scenario design scenario where multiple access faces unexpected congestion, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. Test the proposed change under normal and peak traffic before deployment.

Question 178

178. Rural setup integration scenario where spread spectrum has energy inefficiency, examine system behavior in detail. Develop a robust optimization technique.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 179

179. Critical network analysis scenario where 3G exposes security weaknesses, examine system behavior in detail. Investigate causes and model improvements.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. For a critical environment, give priority to reliable service and secure communication.

Question 180

180. Autonomous environment integration scenario where 3G demonstrates poor QoS, examine system behavior in detail. Identify key factors and propose redesign.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 181

181. Intelligent architecture monitoring scenario where 1G experiences frequent handoff failures, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Test the proposed change under normal and peak traffic before deployment.

Question 182

182. Low-latency infrastructure monitoring scenario where 1G reveals packet loss anomalies, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. Test the proposed change under normal and peak traffic before deployment.

Question 183

183. Dense environment analysis scenario where radio interface experiences frequent handoff failures, examine system behavior in detail. Design an optimized communication approach.

Answer: The radio interface carries control and user data between a mobile device and base station. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Signal strength, fading, obstacles, interference and mobility affect its reliability; improve coverage and link adaptation. Test the proposed change under normal and peak traffic before deployment.

Question 184

184. Rural platform design scenario where multiple access encounters scalability challenges, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: Multiple access lets several users share radio spectrum through frequency, time, code or subcarrier separation. Scalability problems occur when users or traffic exceed radio, processing, signalling or backhaul capacity. Measure the bottleneck, use load balancing and admission control, and expand only the constrained resource. Dynamic scheduling, admission control and fair resource allocation help improve capacity and user performance. In a rural deployment, prioritize coverage continuity and efficient use of limited sites.

Question 185

185. Dynamic infrastructure implementation scenario where 3G has energy inefficiency, examine system behavior in detail. Design an optimized communication approach.

Answer: 3G supports mobile voice and data, but capacity and latency are weaker than newer LTE networks. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Tune radio resources and handovers, and shift heavy traffic to 4G/5G where available. In a dynamic platform, review performance during peak and off-peak traffic.

Question 186

186. Autonomous platform analysis scenario where HSCSD suffers from interference issues, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 187

187. Autonomous environment migration scenario where 1G faces unexpected congestion, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 1G uses analogue signals for voice and has low capacity, limited security and no practical modern data support. Congestion occurs when traffic exceeds available radio or transport capacity, causing delay, packet loss and blocking. Identify overloaded cells or links, balance traffic, schedule resources fairly and add capacity at the measured bottleneck. It is unsuitable for present-day services; migration to a digital network is the effective long-term solution. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 188

188. Wide-area environment monitoring scenario where GPRS suffers from interference issues, examine system behavior in detail. Identify key factors and propose redesign.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. For a wide-area system, verify performance across distant cells and transport links.

Question 189

189. Intelligent environment analysis scenario where GPRS exposes security weaknesses, examine system behavior in detail. Develop a robust optimization technique.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Test the proposed change under normal and peak traffic before deployment.

Question 190

190. Next-gen infrastructure implementation scenario where protocol stack shows high latency variation, examine system behavior in detail. Investigate causes and model improvements.

Answer: The protocol stack organizes communication into physical, link, network, transport and application layers. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Monitor the layers separately because radio errors, congestion or protocol faults can affect end-to-end service. For a next-generation network, ensure the change supports growing traffic and future services.

Question 191

191. Autonomous environment monitoring scenario where 5G suffers from interference issues, examine system behavior in detail. Compare alternatives and select best strategy.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 192

192. Urban system design scenario where 5G suffers from interference issues, examine system behavior in detail. Propose enhancements with justification.

Answer: 5G supports high capacity and low-latency services through advanced radio techniques and flexible network architecture. Interference lowers SINR and may cause retransmissions, lower data rates or dropped connections. Check co-channel and adjacent-channel sources, improve channel planning, coordinate neighbouring cells and apply suitable power control. Use beam management, edge computing and network slicing, while maintaining reliable coverage and mobility. Test the proposed change under normal and peak traffic before deployment.

Question 193

193. High-speed setup migration scenario where GPRS demonstrates poor QoS, examine system behavior in detail. Propose enhancements with justification.

Answer: GPRS adds packet data to GSM by sharing available radio time slots; speed changes with signal quality and network load. Poor QoS appears as low speed, high delay, packet loss or call drops, often due to congestion or weak coverage. Set service priorities, apply QoS scheduling and admission control, improve capacity and monitor network performance. Optimize packet-slot allocation and coding, and use LTE for applications needing higher speed. Since the deployment is high-speed, validate the changes while users move between cells.

Question 194

194. Low-latency scenario integration scenario where spread spectrum has energy inefficiency, examine system behavior in detail. Propose enhancements with justification.

Answer: Spread spectrum spreads a signal over a wider bandwidth using codes or hopping, improving resistance to narrowband interference. Latency variation causes jitter, making voice, video and interactive applications inconsistent. Measure delay and queueing, prioritize real-time traffic with QoS, reduce congestion and use edge services when suitable. Good code planning, synchronization and power control are needed to maintain capacity and avoid near-far effects. Test the proposed change under normal and peak traffic before deployment.

Question 195

195. Hybrid scenario evaluation scenario where 4G exposes security weaknesses, examine system behavior in detail. Propose enhancements with justification.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Security weaknesses can expose subscriber information, signalling or user data to unauthorized access or interception. Use strong authentication and encryption, protect keys, patch network elements and disable obsolete protocols. Use carrier aggregation, load balancing and small cells where demand is high. Because this is a hybrid environment, coordinate access technologies and maintain session continuity.

Question 196

196. Autonomous network analysis scenario where HSCSD reveals packet loss anomalies, examine system behavior in detail. Propose enhancements with justification.

Answer: HSCSD increases GSM circuit-switched data speed by assigning multiple time slots to one user. Packet loss affects voice and video quality and causes retransmissions that can increase delay. Locate the loss point, improve radio quality, prevent queue overflow and use suitable error correction and QoS. Because slots remain reserved during the session, use it only for limited legacy needs and prefer packet-based services for bursty data. For an autonomous network, monitor performance indicators and adjust resources when conditions change.

Question 197

197. Critical environment integration scenario where 2G has energy inefficiency, examine system behavior in detail. Examine trade-offs and recommend solutions.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For a critical environment, give priority to reliable service and secure communication.

Question 198

198. Dynamic platform monitoring scenario where 4G experiences frequent handoff failures, examine system behavior in detail. Analyze the behavior and recommend improvements.

Answer: 4G LTE provides high-speed IP data using efficient radio scheduling, but performance depends on signal quality, cell load and backhaul. Handoff failures interrupt a session when a device moves between cells; late decisions, poor target coverage or missing neighbour data may be responsible. Update neighbour lists, tune handover thresholds and test mobility routes to reduce dropped sessions and unnecessary switching. Use carrier aggregation, load balancing and small cells where demand is high. In a dynamic platform, review performance during peak and off-peak traffic.

Question 199

199. Critical system implementation scenario where 2G has energy inefficiency, examine system behavior in detail. Evaluate system performance and justify solutions.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. For a critical environment, give priority to reliable service and secure communication.

Question 200

200. Advanced network implementation scenario where 2G has energy inefficiency, examine system behavior in detail. Assess reliability and suggest corrections.

Answer: 2G provides digital voice and basic data, but its bandwidth and data rate are limited. Energy inefficiency increases battery drain or network power use through unnecessary signalling, retransmissions and high transmit power. Use sleep modes and discontinuous reception, apply power control and schedule resources efficiently without harming coverage. Improve cell planning and channel allocation for legacy use, and move data-intensive services to 4G/5G. Test the proposed change under normal and peak traffic before deployment.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____